Lecture 1, part 2 of 3

Law of the instrument

"If all you have is a hammer, everything looks like a nail."

Do we really need ML?

Before you reach for a model, ask if the answer is a rule, a lookup, or a regex.

Predictive or generative?

If it is ML, it is still two different jobs.

- **Predictive**: learn a function from your data. Classify, rank, score, forecast. Trained on your labels, scored against ground truth.
- **Generative**: call a foundation model someone else trained. Text, code, images. No labels of your own, and no single right answer to check against.

Before machine learning

Google's *Rules of ML* opens with three rules you apply *before* any model exists.[1]

1. **Don't be afraid to launch without ML.**
2. **Design and implement metrics first.**
3. **Choose ML over a complex heuristic.**

1. Google, *Rules of Machine Learning*, "Before Machine Learning". https://developers.google.com/machine-learning/guides/rules-of-ml ↩

Rule #1: Don't be afraid to launch without ML

"Machine learning is cool, but it requires data."[1]

- A heuristic gets you roughly **half** the gain of a sophisticated model, for a fraction of the effort.
- Rank apps by install rate. Block the publishers you already know are spam. Curate by hand.
- No data yet means no ML yet. Ship something, and let it generate the data.

1. Google, *Rules of Machine Learning*, Rule #1. https://developers.google.com/machine-learning/guides/rules-of-ml ↩

Rule #2: Design and implement metrics first

"Before formalizing what your machine learning system will do, track as much as possible."[1]

– Users grant tracking access more readily early on than late.
– Historical data is only available if you started logging before you needed it.
– Instrumenting metrics shapes the system you end up building.
– You learn what actually moves, versus what you assumed would.

1. Google, *Rules of Machine Learning*, Rule #2. https://developers.google.com/machine-learning/guides/rules-of-ml ↩

Rule #3: Choose ML over a complex heuristic

"A complex heuristic is unmaintainable."[1]

Simple rules ship your first version. But once you have data and a clear objective, and the rules have grown into a thicket of special cases, that is the signal to switch. Models are easier to update and maintain than a hundred hand-tuned conditionals.

1. Google, *Rules of Machine Learning*, Rule #3. https://developers.google.com/machine-learning/guides/rules-of-ml ↩

These rules in 2026

Start at 1:

1. A rule or heuristic (no model at all).
2. Prompt an existing model.
3. Context and retrieval: RAG.
4. Tools and agents: models that can take actions.
5. Model adaptation: adapt the weights only when nothing above works (SFT/RLHF/KTO).

RECAP: LAW OF THE INSTRUMENT

1. Ask whether a rule, a lookup, or a regex answers it first.
2. Predictive and generative are different jobs, with different baselines and different evals.
3. Launch without ML to earn the data. Switch when the heuristic stops being maintainable.
4. Design your metrics before the system, not after it.
5. Climb the ladder one rung at a time.

NEXT: YOU HAVE DECIDED YOU NEED A MODEL. HOW DO YOU DECIDE?